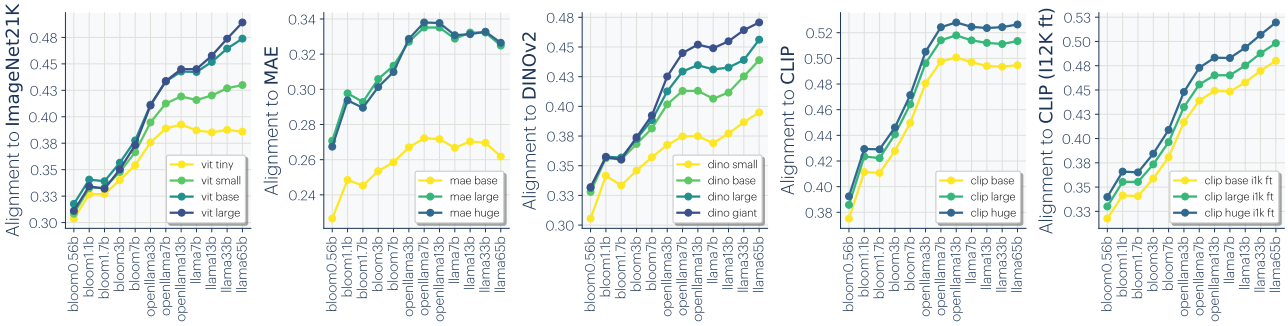
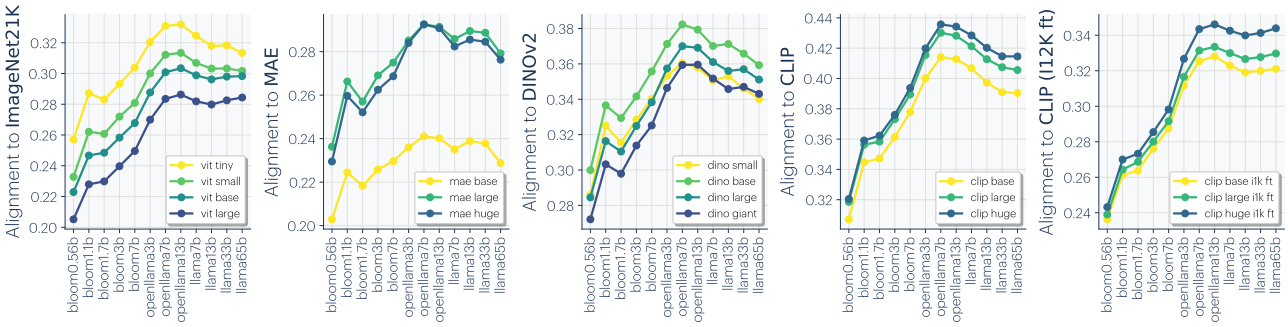
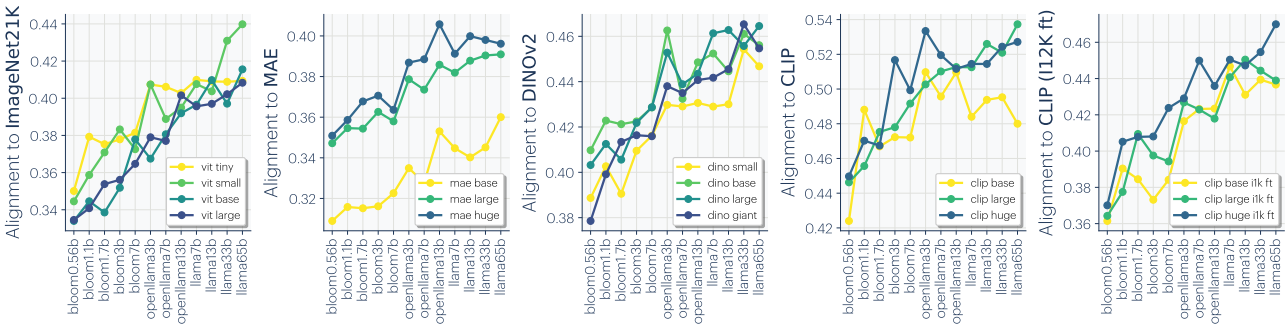


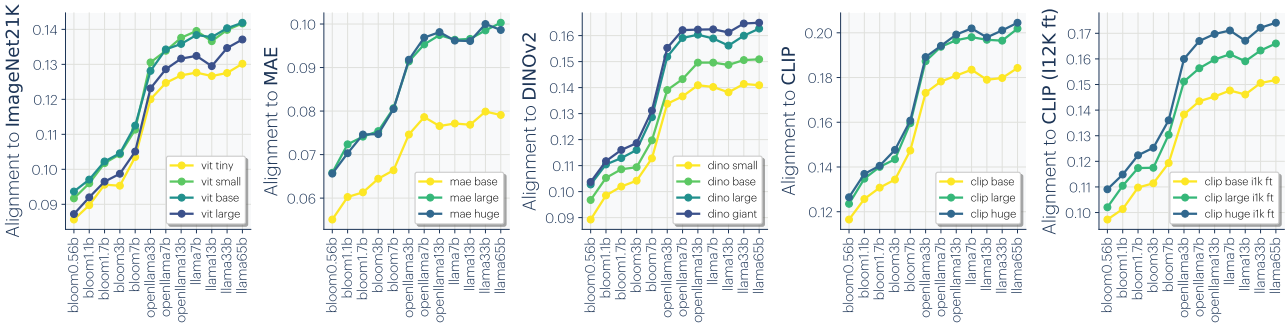
(a) CKA



(b) Unbiased CKA

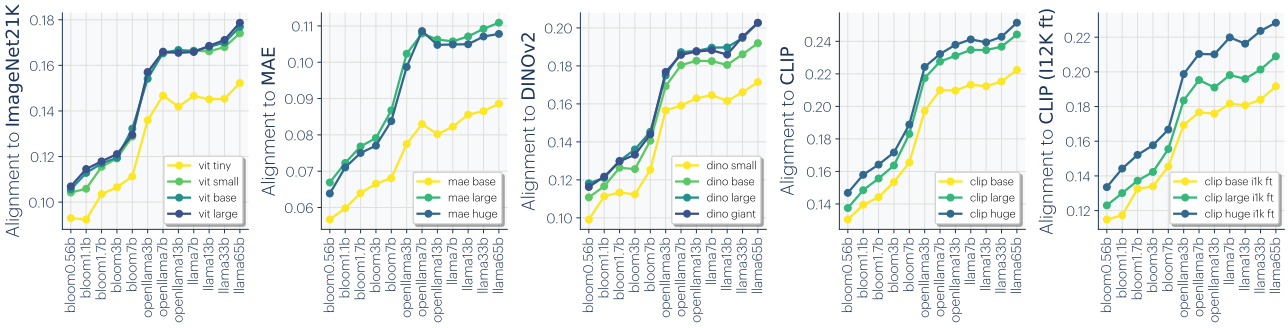


(c) SVCCA

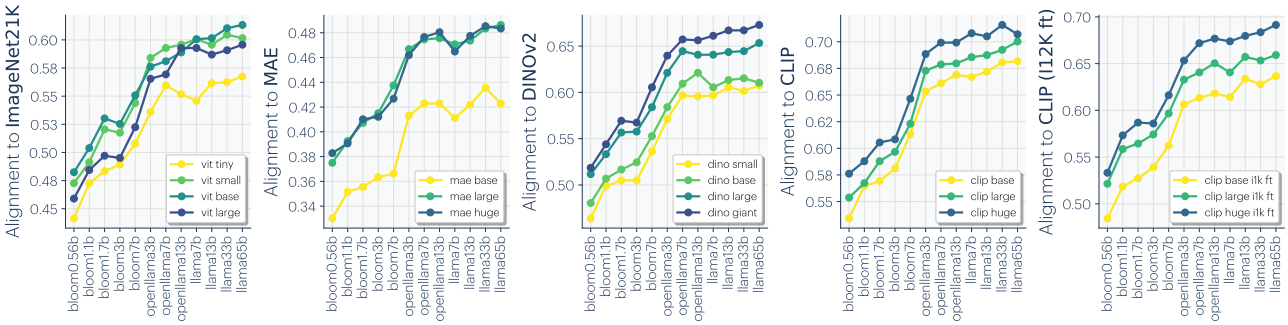


(d) Mutual k-NN ($k = 10$)

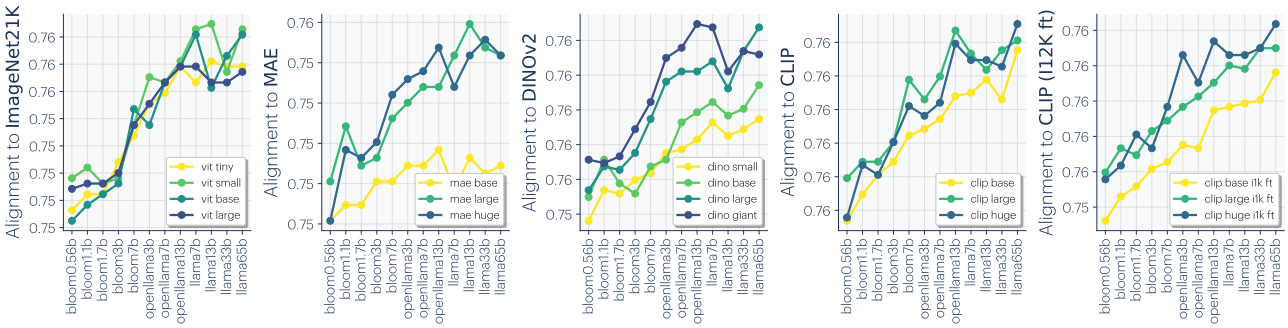
Figure 13. **Cross-modal alignment for various metrics**



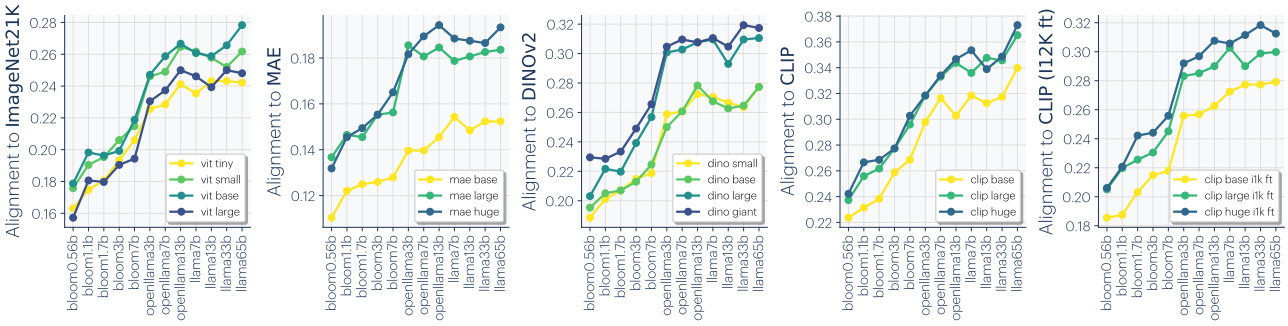
(a) CKNNA ($k = 10$)



(b) Cycle k-NN ($k = 10$)



(c) Edit-distance k-NN ($k = 10$)



(d) Longest-Common-Subsequence k-NN ($k = 10$)

Figure 14. **Cross-modal alignment measured with various metrics**

C. Experiments on Evaluating Alignment and Convergence

To demonstrate representational convergence, we take off-the-shelf models at multiple scales and multiple modalities and measure their representational alignment.

C.1. Vision-Vision Alignment and Representation Quality

We consider 78 vision models in total:

- 17 ViT models ranging from ViT-tiny to ViT-giant, trained on tasks including ImageNet-21k (Dosovitskiy et al., 2020) classification, Masked Autoencoders (He et al., 2021), DINO (Caron et al., 2021), and CLIP (Radford et al., 2021), including some finetuned on ImageNet-12k.
- 1 randomly initialized ResNet-50.
- 11 ResNet-50 models trained with contrastive learning on ImageNet-1k, Places-365 (Zhou et al., 2017; López-Cifuentes et al., 2020), and 9 synthetic image datasets used in Baradad et al. (2022).
- 49 ResNet-18 models trained with Alignment and Uniformity contrastive loss (Wang & Isola, 2020) on ImageNet-100, Places-365, and 47 realistic and synthetic image datasets from Baradad et al. (2021).

To test representation quality, we evaluate linear probing performance on all 19 VTAB classification tasks (Zhai et al., 2019), which is a standard multi-task transfer learning benchmark containing structured, specialized, and natural datasets covering diverse domains. To reduce compute requirements, we subsample training and validation datasets to have at most 10,000 samples. We consider a representation solves a task if its performance is $\geq 80\%$ of the best performance on that task across all 78 models.

To compute the alignment metric, we use $k = 10$ nearest neighbors over 1000 image representations computed on Places-365's validation dataset (Zhou et al., 2017). This dataset is disjoint from VTAB datasets, although both contain natural images.

C.2. Cross-Modal Alignment

We compare the representation of an image in a vision model to the representation of a caption describing that image in a language model. The language model families we consider are BLOOM (BigScience et al., 2022), OpenLLaMA (Geng & Liu, 2023), and LLaMA (Touvron et al., 2023). For Figure 4, we included more recent model families such as OLMo (Groeneveld et al., 2024), LLaMA3 (Meta, 2024), Gemma (Team et al., 2024), and Mistral/Mixtral (Jiang et al., 2023; 2024). These models were downloaded from Huggingface (Wolf et al., 2019).

For vision models, we consider ViT models (Dosovitskiy et al., 2020) of various sizes trained on various data and objectives. We mainly consider the popular vision models: classification on ImageNet-21K (Russakovsky et al., 2015), MAE (He et al., 2021), DINOv2 (Oquab et al., 2023), CLIP (Radford et al., 2021), and CLIP finetuned on ImageNet-12K. These models were downloaded from PyTorch Image Models (TIMM; Wightman (2021)). This is a subset of the models used in vision-vision comparison.

To compute the alignment metric, we use $k = 10$ nearest neighbors over 1024 samples from WIT (Wikipedia-based Image Text; Srinivasan et al. (2021)). For the vision model, we use class token of each layer, and for the language model, we average pool each layer to a single token. Since it is not trivial to determine where the alignment might occur, we draw inspiration from BrainScore(Schrimpf et al., 2018) and compute pairwise alignment scores, then take the maximum. One of these pairwise comparisons also includes concatenated features. We apply l_2 normalization to the features before measuring the distance. As transformer architectures have "emergent outliers" (Dettmers et al., 2022), we truncate the elements in the features that are above the 95-th percentile.

Simply taking the last token did not show any strong alignment signal. We also experimented with prompting the language model and taking the last token representation. The prompt we used was

```
An image with the caption '<caption>'. This is an image of a <fill>
```

Using prompting showed similar trends to average pooling but had slightly lower alignment scores.